Engineering Team Charter

1. Team Name

EMG Acquisition Team - MyoMetrics

2. Mission Statement

To acquire clear, consistent EMG signals with the MyoWare 2.0 sensor, develop MATLAB tools that filter, rectify and analyse muscle activation, and show how contraction conditions shape the signal, while every member contributes through shared responsibility and accountable teamwork.

3. Goals & Objectives

- Goal 1: We will successfully acquire EMG signals using the MyoWare 2.0 muscle sensor from at least one participant, with additional participants where equipment and time permit, by the end of the first project week, ensuring that the signals are sufficiently clear and consistent for further analysis
- Goal 2: We will develop a MATLAB script capable of importing and processing EMG signals, implementing appropriate filtering and rectification techniques to reduce noise and identify muscle activation levels by the end of the project.
- Goal 3: We will investigate the relationship between muscle activation and EMG signal characteristics by recording and comparing signals under at least three different muscle contraction conditions by the end of the project.
- Goal 4: We will ensure that all group members contribute to the project by allocating individual responsibilities, regularly reviewing our progress and completing agreed tasks within the project deadlines.

4. Team Members & Roles

Name	Role	Responsibilities
Tariq El-Jumaily	Project Lead	Coordinates team activities, delegates tasks, monitors deadlines and oversees project delivery.
Olly Gimblett	Signal Processing Engineer	Develops MATLAB algorithms for EMG filtering, noise reduction and signal visualisation.

Name	Role	Responsibilities
Aranan Theivendra	Data Acquisition Engineer	Manages EMG hardware setup, electrode placement, data collection and signal integrity.
Shiv Desai	Data Analysis Engineer	Develops feature extraction algorithms to identify muscle activation, contraction patterns and fatigue indicators.
Thomas Wood	Quality & Documentation Lead	Manages quality assurance, technical documentation, testing records and audit preparation.

5. Resources & Constraints

Resources

- Group of 5 people
- Lab demonstrators to help with readings and work
- Software; MATLAB, Github, Google Docs, Google Slides
- EMG Hardware; Myoware 2.0

Constraints

- One ECG/EMG kit per 2 groups
- Pick up Tue AM 03.11
- 5 days to complete project
- QMS Audit on Thu 05.11
- Video Submission Thu 05.11 5PM
- Kit Return Fri 06.11

6. Operating Norms

- **Daily meeting at 9:15 in the labs:** Each member will state yesterday's work and output, today's tasks and any other issues with the work in general. This will last around 10-15 minutes.
- **End-of-day check-in:** 16:00 to confirm the day's criteria is met and all raw data is backed up.
- Communication: Microsoft Teams group chat during lab hours to record, share and save all data. Agreements within the team will be made here.
- Documentation: All files to be shared on the Github repository and named systematically by task and version. Raw data is never overwritten or edited
- Availability: Messages answered within 2 hours during lab hours; absences flagged to the team before 09:00

7. Decision-Making Process

- **Technical decisions:** Decisions will be made collaboratively, considering experimental evidence, feasibility and project objectives. Where consensus cannot be reached, a majority vote will be held, with the Team Lead acting as tie-breaker.
- **Project Management Decisions:** The Team Lead will coordinate task allocation, priorities and deadlines, consulting relevant team members before making significant changes.
- **Equipment & Experimental Decisions:** Decisions regarding EMG acquisition, electrode placement and testing procedures will be agreed with the partner team, following laboratory guidance.
- **Urgent Decisions:** Time-sensitive issues will be addressed by the Team Lead in consultation with the relevant technical lead. Any changes will be communicated to the team and documented.
- **Laboratory Safety Decisions:** All decisions must comply with laboratory safety requirements and demonstrator guidance.

8. Conflict Resolution

- The conflict will be raised directly and privately between the members involved, focused on the work and the issues at hand rather than the person.
- If unresolved, the Project Lead will mediate privately and promptly, with unresolved technical disagreements escalated to a supervisor where appropriate.
- Underperformance will be addressed the same day by reallocating tasks and agreeing a catch-up deadline, this will be recorded on the Teams chat.
- All disagreements handled with civility: we challenge the idea, not the individual, and the decision stands once made together.

9. Team Values & Culture

Openness: Results, including failed recordings and unexpected data, are shared freely with the whole team

Ongoing refinement: Each recording session ends with a short review of what to change next time

Inclusivity of thought: Every member's view is heard before a decision is made, regardless of seniority or experience

Ownership: Members take responsibility for their tasks and report progress honestly, especially when something goes wrong

Constructive debate: Challenge ideas, not people, and keep discussion focused on evidence such as signal quality and filter choices

10. Sign-Off & Commitment

Name	**Signature**	**Date**
Tariq El-Jumaily		02/11/2026
Olly Gimblett		02/11/2026
Aranan Theivendra		02/11/2026
Shiv Desai		02/11/2026
Thomas Wood		02/11/2026